DATABASE PROJECT

Wadi Watch

Flash Flood Early Warning and Response System

Entities and Attributes

This table lists every entity in the database with its attributes. Each attribute is marked as a primary key, a foreign key or a simple attribute. Sensors has one composite attribute (Valley).

ENTITY	ATTRIBUTE	TYPE
Town	town_ID	**Primary Key**
	town_Name	simple
	Location	simple
	Current_Risk_Level	simple
Sensors	sensor_ID	**Primary Key**
	sensor_Type	simple
	sensor_Location	simple
	Status	simple
	last_Reading	simple
	Valley	composite (valley_ID, Name, Boundary)
Team	team_ID	**Primary Key**
	team_Name	simple
	team_Location	simple
	Status	simple
	workload	simple
Road	area_ID	**Primary Key**
	area_Name	simple
	Location	simple
	Current_Risk_Level	simple
Citizen	citizen_ID	**Primary Key**
	citizen_Name	simple
	phone_Number	simple

ENTITY	ATTRIBUTE	TYPE
Alert	alert_ID	**Primary Key**
	town_ID	**Foreign Key**
	road_ID	**Foreign Key**
	alert_Level	simple
	time_Raised	simple
	decision	simple
	confirmed_by	**Foreign Key**
	decision_Time	simple
Warning	Warning_ID	**Primary Key**
	Alert_ID	**Foreign Key**
	Message_Text	simple
	Language	simple
	Channel	simple
	Delivery_Count	simple
Citizen Report	Report_ID	**Primary Key**
	Reporter_ID	**Foreign Key**
	Location	simple
	Photo	simple
	Report_Type	simple
	Urgency	simple
	Submission_Time	simple
	Submission_Channel	simple
Supply	Supply_ID	**Primary Key**
	Supply_Name	simple
	Quantity	simple
	Shelter_ID	**Foreign Key**
Shelter	Shelter_ID	**Primary Key**
	Shelter_Name	simple
	Location	simple
	Capacity	simple
	Current_Occupancy	simple
	Occupancy_Percentage	simple
Equipment	Equipment_ID	**Primary Key**
	Equipment_Type	simple
	Equipment_Status	simple
	Quantity	simple

ENTITY	ATTRIBUTE	TYPE
Shelter Registration	Registration_ID	**Primary Key**
	Shelter_ID	**Foreign Key**
	Person_ID	**Foreign Key**
	Arrival_Time	simple
	Family_ID	**Foreign Key**
	Volunteer_ID	**Foreign Key**
Volunteer	Volunteer_ID	**Primary Key**
	Volunteer_Name	simple
	Phone_Number	simple
Person	person_ID	**Primary Key**
	Name	simple
	Phone_Number	simple
Risk Assessment	Assessment_ID	**Primary Key**
	Town_ID	**Foreign Key**
	Road_ID	**Foreign Key**
	Assessment_Time	simple
	Risk_Level	simple
	Confidence_Level	simple
Weather Forecast	Forecast_ID	**Primary Key**
	Forecast_Time	simple
	Rainfall_Prediction	simple
	Location	simple
	Source	simple
Flood Event	Event_ID	**Primary Key**
	Event_Start_Time	simple
	Event_End_Time	simple
	Location	simple
	Event_Status	simple
Event Report	Event_Report_ID	**Primary Key**
	Event_ID	**Foreign Key**
	Warnings_Count	simple
	Response_Time	simple
	People_Rescued	simple
	Report_Date	simple
Officer	Officer_ID	**Primary Key**
	Officer_Name	simple
	Role	simple
	Phone_Number	simple

ENTITY	ATTRIBUTE	TYPE
Task Status Update	Update_ID	**Primary Key**
	Task_ID	**Foreign Key**
	Status	simple
	Update_Time	simple
	Description	simple
Family	Family_ID	**Primary Key**
	Family_Name	simple
	Family_Size	simple
	Contact_Number	simple
Rescue Task	Task_ID	**Primary Key**
	Task_Location	simple
	Priority	simple
	Task_Status	simple
	Team_ID	**Foreign Key**
	Officer_ID	**Foreign Key**
	Report_ID	**Foreign Key**

Relationships

This table shows how the entities are connected, with the cardinality and the participation of each relationship (single = partial, double = total).

RELATIONSHIP	ENTITY 1	ENTITY 2	CARDINALITY	PARTICIPATION
Produces	Sensor	Reading	1:M	Single / Double
Generates	Alert	Warning	1:M	Single / Double
Has	Town	Alert	1:M	Single / Double
Affects	Alert	Road	M:N	Single / Single
Includes	Flood Event	Alert	1:M	Single / Double
Confirms	Officer	Alert	1:M	Single / Single
Produces	Flood Event	Event Report	1:M	Single / Double
Contributes to	Weather Forecast	Risk Assessment	1:M	Single / Double
Indicates	Risk Assessment	Road	M:N	Single / Single
Has	Shelter	Supply	1:M	Single / Double
Belongs to	Equipment	Team	M:N	Single / Single
Has	Shelter	Shelter Registration	1:M	Single / Double
Includes	Shelter Registration	Volunteer	M:N	Single / Single
Registers	Shelter Registration	Person	1:M	Single / Double
Has	Person	Family	1:M	Single / Single
Has	Rescue Task	Task Status Update	1:M	Single / Double
Handles	Rescue Team	Rescue Team	1:M	Single / Double
Receives	Rescue Team	Rescue Task	1:M	Single / Single
Generates	Citizen	Citizen Report	1:M	Single / Double